Laboratory 3:

Signals and Systems
CT/DT Systems and Basic System Properties

System Properties Explorer: Hardware Verification

Lab Duration: 2 hours
Pre-Lab Time Expected: 45 minutes (property predictions, test signal design, MATLAB plan)
Key Concept: Mapping formal system property definitions to measurable hardware behavior and quantifying agreement with rigor sufficient for production certification.

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Grading and Authenticity Requirements

Deliverable (1/13): This is a graded laboratory assignment. Include your **name and student ID** in all required screenshots and photographs (oscilloscope captures, MATLAB plots, breadboard/setup photos).

Deliverable (2/13): All MATLAB property verification must use your own captured input/output data from hardware. Purely synthetic data generated in MATLAB is not accepted.

Checkpoint (1/8): Before submission, verify that all screenshots are legible, include axis labels with units, and clearly show measurement readouts where applicable.

Title & Real-World Context

You are a Junior RF Circuit Engineer at a 5G Base Station Equipment Manufacturer

Your technical lead has tasked you with validating the analog front-end signal conditioning chain for a next-generation **5G New Radio receiver module** deployed in carrier networks worldwide. The receiver must reliably handle three critical operational scenarios:

1. **Wideband Receiver Linear Response (Primary Requirement).** Your module receives 5G signals across a 100+ MHz bandwidth with peak-to-peak amplitudes ranging over 60 dB (from barely detectable μV signals to interference transients at 1V+). A **first-order RC low-pass filter** ahead of the ADC must suppress out-of-band interference while remaining perfectly **linear** in the passband (any nonlinearity causes IMD—intermodulation distortion—that corrupts adjacent-channel decoding). The receiver's linearity specification: $< 0.5\%$ THD across the entire operating range. A single component mismatch or thermal drift that violates this linearity destroys the receiver's ability to decode weak signals amid interference, causing field deployment failures and customer complaints.
2. **Envelope Detection for Signal Power Measurement (Automatic Gain Control).** A **diode clipper** detects signal envelopes for automatic gain control (AGC). This circuit must respond proportionally to input amplitude (linearity requirement) and show consistent time-delayed response to amplitude changes (time-invariance requirement). If the clipper is nonlinear or its threshold drifts with temperature, the AGC loop becomes unstable and oscillates, creating audible noise and dropped calls. Your lab validates both linearity and time invariance under realistic load and temperature conditions.
3. **Threshold Detection and Alarm Generation (System Stability Certification).** A **comparator threshold detector** flags excessive interference or system overload. This circuit must be stable (bounded output for bounded input, no ringing or oscillation) and causal (output does not respond before the input edge arrives). If the comparator rings or exhibits instability, it generates false alarms that crash the receiver or trigger unnecessary power backoffs, degrading cell-site throughput.

Your Mission. In this lab, you will systematically test three fundamental system properties—linearity, time invariance, causality, and BIBO stability—on real circuits and defend your findings with both mathematical reasoning and quantified hardware evidence:

- Design unique, reproducible test signals using your TA-assigned random seed.
- Implement hardware property tests: superposition for linearity, delayed inputs for time invariance, edge detection for causality, amplitude sweeps for stability.
- Acquire dual-channel oscilloscope and serial MATLAB data streams.
- Quantify agreement between prediction and measurement using normalized RMS error.
- Identify failure mechanisms when properties are violated (saturation, threshold, settling time, etc.).

The telecommunications industry certifies base station equipment to 3GPP/ITU standards that require formal property verification. A receiver that appears to work in the lab but exhibits hidden nonlinearity, time variance, or instability in the field can render an entire cell site inoperable under peak traffic. Your systematic validation approach is the engineering discipline that separates prototype breadboards from production equipment.

Constraints (Fixed). 2-hour lab execution window, existing hardware platform (Arduino Uno, oscilloscope, MATLAB). Unique TA-assigned test parameters ensure independent work. All skeleton code provided. All checkpoints and deliverables remain exactly as specified herein.

Upon completion of this laboratory, you will demonstrate:

1. **Rigorous System Property Definition and Measurement.** State mathematical definitions of linearity, time invariance, causality, and BIBO stability, and design repeatable experiments that test each property under controlled conditions with quantifiable tolerances.

2. **Hardware-Software Property Validation.** Acquire dual-channel oscilloscope and MATLAB serial data, align time vectors, and compute superposition residuals and time-shift errors with sufficient precision to distinguish real property violations from noise and measurement artifacts.

3. **Root-Cause Analysis of Property Failures.** When linearity or time invariance fails, identify and explain the underlying physical mechanism (saturation, threshold nonlinearity, settling delay, finite slew rate, or hysteresis) rather than simply reporting a measurement.

4. **Quantitative Error Metrics and Production-Grade Acceptance Criteria.** Calculate normalized RMS error (NRMSE) between predicted and measured responses, set acceptance thresholds (e.g., $< 2\%$ for linearity), and justify criteria by reference to system-level performance requirements.

5. **Causality and Stability Certification via Edge-Detection and Amplitude-Sweep Tests.** Verify that output responses do not precede input changes (causality) and that bounded inputs produce bounded outputs without ringing (BIBO stability), using oscilloscope time-domain evidence and MATLAB statistical analysis.

6. **Systematic Documentation and Defense Methodology.** Prepare a formal property truth table with both qualitative justifications (circuit architecture, component values) and quantitative evidence (measured data, error metrics, screenshots) suitable for presentation to design review boards.

1 Core Concepts: System Properties and Measurement Methodology

For a system operator $\mathcal{T}\{\cdot\}$ with input $x(t)$ and output $y(t)$, four fundamental properties govern its behavior and suitability for signal processing applications:

- **Linearity:**
$$\mathcal{T}\{ax_1(t) + bx_2(t)\} = a\,\mathcal{T}\{x_1(t)\} + b\,\mathcal{T}\{x_2(t)\}$$
for all scalars a, b and admissible inputs x_1, x_2.
- **Time Invariance:**
$$x_\Delta(t) = x(t - \Delta) \Rightarrow \mathcal{T}\{x_\Delta(t)\} = y(t - \Delta)$$
where $y(t) = \mathcal{T}\{x(t)\}$.
- **Causality:** $y(t_0)$ depends only on input values at times $\tau \le t_0$.
- **BIBO Stability:**
$$|x(t)| \le M_x < \infty \Rightarrow |y(t)| \le M_y < \infty$$

For sampled data $x[n], y[n]$, use discrete analogs. To quantify test agreement, use normalized RMS error:
$$\mathrm{NRMSE}(\%) = 100 \times \frac{\|y_{test}[n] - y_{ref}[n]\|_2}{\|y_{ref}[n]\|_2}$$
Small NRMSE supports property agreement over the tested operating region; it does not prove global validity.

2 Pre-Laboratory Assignment

Pre-lab Deliverable (1/2): Complete all pre-lab work before arriving. Bring handwritten derivations and predicted truth table.

2.1 Part A: Property Predictions

For each system below, predict whether it is linear, time-invariant, causal, and BIBO stable. Justify each claim in 2–4 technical sentences.

1. RC low-pass filter (first-order passive network)
2. Diode clipper (single-supply threshold clipping)
3. Comparator (open-loop threshold detector)

2.2 Part B: Test Signal Design

Pre-lab Deliverable (2/2): Create your own unique test set using your assigned random seed.Design and tabulate:

- Two base signals $x_1(t), x_2(t)$ (distinct shapes/frequencies)
- Scalars a, b for superposition test
- Time shift Δ (ms)
- Bounded amplitude limits for stability tests

2.3 Part C: MATLAB Verification Plan

Write pseudocode for:

1. importing captured CSV/serial data,
2. aligning time vectors and resampling if needed,
3. computing superposition residual and NRMSE,
4. computing time-invariance residual and NRMSE.

Deliverable (3/13): Submit scanned handwritten pre-lab pages with your name and student ID visible on the first page.

3 Equipment and Setup

Table 1. Required Hardware and Software

Category	Required Items
Hardware Source	Arduino (Uno/Mega or approved equivalent), USB cable, breadboard, jumper wires
Circuit Components	Resistors, capacitors, signal diode (e.g., 1N4148), op-amp/comparator IC, potentiometer (threshold set)
Measurement	Oscilloscope or ADALM2000, multimeter, probes/ground clips
Software	Arduino IDE, MATLAB (Signal Processing and basic plotting capability)
Documentation	Lab notebook pages for handwritten property truth tables

3.1 Assigned Parameters

Table 2. TA-Assigned Experiment Parameters

Symbol	Description	Assigned Value
f_1	Frequency for x_1	______________
f_2	Frequency for x_2	______________
a, b	Superposition coefficients	______________
Δ	Time shift for TI test	______________
A_{max}	Maximum allowed input amplitude	______________
Seed	Unique randomization seed	______________

Deliverable (4/13): Capture one physical setup photo that includes all three circuits (or three separate photos), each with your student ID visible in-frame.

4 Laboratory Procedure

4.1 Phase 1: Circuit Construction and Baseline Validation

4.1.1 Step 1.1 RC Filter Assembly

Checkpoint (2/8): Build and verify a first-order RC low-pass circuit driven by Arduino output.

Procedure:

1. Construct RC filter and verify wiring against your schematic.
2. Apply sinusoidal input and capture $x(t)$ and $y(t)$ on two channels.
3. Record gain and phase shift at your assigned frequency.

Deliverable (5/13): Submit dual-channel scope screenshot showing input/output waveforms with measured amplitude ratio and time delay annotations.

Step 1.1 Evidence: RC filter scope capture with ID visible

4.1.2 Step 1.2 Diode Clipper Assembly

Checkpoint (3/8): Build the clipper and demonstrate amplitude limiting behavior.

Procedure:

1. Build a diode clipping circuit for positive or negative threshold clipping.
2. Sweep input amplitude while observing output flattening/clipping onset.
3. Record approximate clipping threshold from measurement.

Deliverable (6/13): Submit scope screenshot with clipped waveform and labeled threshold estimate.

Step 1.2 Evidence: Diode clipper scope capture with ID visible

4.1.3 Step 1.3 Comparator Circuit Assembly

Checkpoint (4/8): Build comparator with adjustable threshold and verify binary switching behavior.

Procedure:

1. Set threshold using potentiometer or reference divider.
2. Apply analog input and observe output transitions.
3. Measure switching threshold and discuss hysteresis (if present).

Deliverable (7/13): Submit scope screenshot showing analog input and digital-like comparator output with threshold annotation.

Step 1.3 Evidence: Comparator scope capture with ID visible

4.2 Phase 2: Property Testing with Hardware Data

4.2.1 Step 2.1 Linearity Test (Superposition)

Checkpoint (5/8): For each circuit, acquire data to compare $\mathcal{T}\{ax_1 + bx_2\}$ versus $a\mathcal{T}\{x_1\} + b\mathcal{T}\{x_2\}$.

Required acquisition set per circuit:

1. Input/output for x_1
2. Input/output for x_2
3. Input/output for $ax_1 + bx_2$

Deliverable (8/13): Provide MATLAB plot overlay and NRMSE table quantifying superposition agreement for all three circuits.

Step 2.1 Evidence: MATLAB linearity overlays and NRMSE summary

Discussion (1/7): Explain why linearity is expected to hold (or fail) for each circuit. Distinguish intrinsic circuit behavior from measurement noise.

4.2.2 Step 2.2 Time-Invariance Test

Checkpoint (6/8): For each circuit, compare response to shifted input with shifted response to original input.

Required acquisition set per circuit:

1. Input/output for $x(t)$
2. Input/output for $x(t - \Delta)$

Deliverable (9/13): Provide MATLAB overlay of $\mathcal{T}\{x(t-\Delta)\}$ and $y(t-\Delta)$ and report NRMSE for each circuit.

Step 2.2 Evidence: MATLAB TI overlays and NRMSE summary

Discussion (2/7): If a mismatch appears, determine whether it is from non-TI behavior, trigger alignment error, finite sampling effects, or threshold drift.

4.2.3 Step 2.3 Causality and Stability Checks

Checkpoint (7/8): Execute bounded test inputs and confirm no anticipatory output behavior.

Procedure:

1. Apply bounded step/chirp/sinusoid inputs within $\pm A_{max}$.
2. Verify output remains bounded and note any saturation limits.
3. Use a delayed input onset and inspect output before onset to assess causality.

Deliverable (10/13): Submit scope evidence for causality check and bounded-output behavior for each circuit.

Step 2.3 Evidence: Scope captures for causality and BIBO checks

Discussion (3/7): State practical limits of your conclusions: tested amplitude range, bandwidth, and operating region.

4.3 Phase 3: MATLAB Verification from Captured Data

Checkpoint (8/8): Run your verification scripts using only acquired hardware data files.

4.3.1 Step 3.1 Data Import and Alignment

Deliverable (11/13): Submit MATLAB code snippets that import your captured datasets and align time bases for fair comparison.

Step 3.1 Evidence: MATLAB import/alignment code screenshot

4.3.2 Step 3.2 Numerical Property Metrics

Deliverable (12/13): Compute and report per-circuit NRMSE for linearity and time-invariance tests.

Table 3. Numerical Verification Summary

Circuit	Linearity NRMSE (%)	TI NRMSE (%)
RC filter	________________	________________
Diode clipper	________________	________________
Comparator	________________	________________

4.3.3 Step 3.3 Property Truth Table

Deliverable (13/13): Complete a handwritten truth table (Linear/TI/Causal/Stable) for each circuit and upload a scan/photo with your ID visible.

Table 4. System Property Truth Table (Final Conclusions)

System	Linear	TI	Causal	BIBO Stable	One-Sentence Justification
RC filter					
Diode clipper					
Comparator					

Step 3.3 Evidence: Handwritten truth table image with student ID

5 Discussion and Interpretation

Discussion (4/7): For each circuit, explain the dominant mechanism that determines property outcomes (memory, thresholding, clipping, or switching).

Discussion (5/7): Why can a system be causal and stable but still non-linear? Provide one example from your measured circuits.

Discussion (6/7): Describe one case where numerical verification might falsely suggest a property due to poor synchronization or insufficient input diversity.

Discussion (7/7): Connect your findings to model validity: when is an LTI approximation useful, and when does it fail for real hardware?

6 Final Submission Checklist

- ☐ Physical circuit photo(s) with student ID visible.
- ☐ Unique test input definitions (with assigned seed values).
- ☐ Scope evidence for RC, clipper, comparator baseline behavior.
- ☐ MATLAB linearity verification plots and NRMSE values.
- ☐ MATLAB time-invariance verification plots and NRMSE values.
- ☐ Causality and stability evidence captures.
- ☐ Handwritten property truth table with justifications and scope references.
- ☐ MATLAB scripts used for verification with captured data file references.

7 Grading Rubric (100 points)

Table 5. Lab 03 Evaluation Rubric

Category	Points	Criteria
Pre-lab quality	15	Correct predictions, clear derivations, and feasible test design.
Circuit implementation and evidence	20	Functional RC/clipper/comparator with clear annotated measurements.
Linearity verification	15	Proper superposition experiments with quantitative error analysis.
Time-invariance verification	15	Shift-based tests with correct alignment and numerical support.
Causality and stability evaluation	15	Convincing bounded-input and no-anticipation evidence.
Truth table and technical justification	10	Handwritten table accurate and defensible with data links.
Code quality and data authenticity	10	MATLAB uses captured data, reproducible workflow, readable scripts.

References and Inspiration

This laboratory explores fundamental system properties that are central to signals and systems education across leading universities.

University Course Inspirations

MIT 6.003: Signals, Systems, and Transforms

Lab 03 is directly inspired by MIT's classical treatment of system properties (linearity, time-invariance, causality, BIBO stability, memorylessness). MIT's approach to characterizing systems through input-output behavior and verifying properties experimentally informs this lab's methodology of testing systems with carefully chosen inputs (impulses, steps, sinusoids) and observing outputs.

MIT 6.007: Applied Electromagnetism and Circuits

The hardware implementation of systems (RC circuits, op-amp circuits, filters) and the experimental verification of their properties reflect MIT 6.007's emphasis on understanding circuits as systems with definable input-output relationships and inherent properties.

Georgia Tech ECE 2026: Real-Time System Characterization

Georgia Tech's approach to characterizing system response in real time—using oscilloscope measurements of transient and steady-state behavior—informs this lab's emphasis on direct hardware observation rather than relying solely on simulation.

Rice University ELEC 241: Embedded System Analysis

The focus on testing system properties under real-time constraints and understanding how non-idealities (op-amp slew rate, finite bandwidth, quantization) affect system characterization reflects Rice's practical approach to embedded systems.

Duke University ECE 280: Lab Modernization

This lab is part of Duke's curriculum redesign emphasizing the connection between abstract system properties and concrete, observable hardware behavior.

References

- A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed. Pearson, 1997. Chapter 2: Linear Time-Invariant Systems.
- A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010. Chapter 2: Discrete-Time Systems.
- J. H. McClellan, R. W. Schafer, and M. A. Yoder, *DSP First: A Multimedia Approach*, 2nd ed. Pearson, 2015. Chapters on system properties and linearity.
- Texas Instruments Op-Amp Applications Handbook (circuits and practical system characterization).